



**EMBEDDED SYSTEM FINAL PROJECT REPORT
DEPARTMENT OF ELECTRICAL ENGINEERING
UNIVERSITAS INDONESIA**

The DuckSaver (An Arduino based Turbidity Sensor System)

GROUP 5

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PREFACE

With deep gratitude, we would like to express our praise and thanks to Almighty God for His blessings and grace, which enabled us to complete the final project for the Embedded Systems course entitled “The DuckSaver (An Arduino Based Turbidity Sensor System)”.

We would like to show appreciation to Mr. Fransiskus Astha Ekadiyanto, the lecturer of the Embedded Systems course, for the valuable knowledge provided throughout the semester. We also wish to thank all the assistants of the Digital Laboratory for their technical support and assistance for the laboratories. Particularly, we wish to express our gratitude to our assistant for the final project, Christian Hadiwijaya, for his contribution and guidance during the project implementation.

As this project is part of our learning process, we recognize that it is not optimized and it can have flaws. Therefore, we are willing to accept any constructive criticism and suggestions for further improvements.

We hope that this project can provide benefits to the reader and become a positive contribution to the development of learning in the Embedded Systems course.

Depok, May 17, 2026

Group 5

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION

- 1.1 Background
- 1.2 Project Description
- 1.3 Objectives
- 1.4 Roles and Responsibilities

CHAPTER 2: IMPLEMENTATION

- 2.1 Equipment
- 2.2 Implementation

CHAPTER 3: TESTING AND ANALYSIS

- 3.1 Testing
- 3.2 Result
- 3.3 Analysis

CHAPTER 4: CONCLUSION

REFERENCES

APPENDICES

- Appendix A: Project Schematic
- Appendix B: Documentation

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Water turbidity is an important environmental indicator that reflects the clarity and quality of water. High turbidity levels are generally caused by suspended particles such as mud, organic matter, algae, and microorganisms. Excessive turbidity can significantly disturb aquatic ecosystems, particularly affecting wildlife such as ducks and other water birds.

One of the major consequences of turbidity is the disappearance of food resources. Turbid water blocks sunlight penetration, preventing the growth of submerged aquatic plants (macrophytes), which are essential habitats and food sources for many aquatic organisms consumed by ducks.

In addition, reduced sunlight limits photosynthesis and decreases dissolved oxygen levels in water. This can create anaerobic conditions that may lead to fish mortality and overall ecosystem imbalance.

Turbid water also reduces underwater visibility, making it more difficult for ducks to detect prey and identify potential predators. Furthermore, high turbidity is often associated with increased concentrations of organic matter and harmful microorganisms, which may expose birds to diseases, toxic algae, or skin infections.

For these reasons, monitoring water turbidity is essential for protecting aquatic biodiversity and maintaining healthy ecosystems.

1.2 PROJECT DESCRIPTION

The DuckSaver project is an automated water quality and turbidity monitoring system designed to analyze water clarity in real time using an analog turbidity sensor and an ATmega328P microcontroller.

The system is programmed entirely in AVR Assembly language and integrates several embedded system concepts such as analog-to-digital conversion (ADC), timer interrupts, UART communication, and I2C-based LCD interfacing.

The device continuously measures water turbidity levels and determines whether the water condition is considered clean or dirty based on a predefined threshold value. The results are displayed through visual indicators such as an LCD screen and LED alerts, while status information is also transmitted through UART serial communication for monitoring purposes.

The project aims to provide a simple, efficient, and low-level embedded solution for environmental monitoring applications.

1.3 OBJECTIVES

The objectives of The DuckSaver project are:

1. To design an embedded system capable of monitoring water turbidity in real time.
2. To detect unsafe water conditions using an analog turbidity sensor and ADC processing.
3. To provide immediate visual feedback through LEDs and an LCD display.
4. To implement serial communication for telemetry and monitoring.
5. To apply low-level AVR Assembly programming concepts in a practical environmental monitoring project.
6. To raise awareness about the environmental impact of water pollution on ducks and aquatic ecosystems.

1.4 ROLES AND RESPONSIBILITIES

The roles and responsibilities assigned to the group members are as follows:

Roles	Responsibilities	Person in charge
Hardware design & Circuit assembly	Designing and assembling the electronic circuit and sensor integration	Syifa Sarah Nuraini Nadira Fayyaza Aisy
Code development	Developing AVR Assembly code, ADC processing, timer interrupts, UART, and I2C communication	Syifa Sarah Nuraini Nadira Fayyaza Aisy

System testing & Debugging	Performing hardware and software testing, troubleshooting, and calibration	Syifa Sarah Nuraini Nadira Fayyaza Aisy
Documentation & Presentation	Writing the report and preparing the final presentation materials	Abdoul-Vahid Akhmadov Théodène Foulon

Table 1. Roles and Responsibilities

CHAPTER 2

IMPLEMENTATION

2.1 EQUIPMENT

The following components were used to implement The DuckSaver system:

- ATmega328P MCU (Arduino Uno) : Main microcontroller and processing unit
- Analog Turbidity Sensor + Probe : Sensor used to measure water clarity
- 16x2 LCD with I2C Backpack : Display module for system status
- Status LED : Visual alert indicator
- Breadboard & Jumper Wires : Circuit prototyping and connections
- 9V Battery / DC Power Supply : Power source for the system

2.2 IMPLEMENTATION

The DuckSaver system was implemented using the ATmega328P microcontroller programmed entirely in AVR Assembly language. The project focuses on low-level embedded system design, combining hardware interrupts, ADC processing, mathematical computations, UART communication, and direct LCD control into a standalone environmental monitoring platform.

The analog turbidity sensor was connected to pin PC0 (A0) to continuously sample water clarity levels. The sensor outputs an analog voltage proportional to the amount of suspended particles present in the water. This analog signal is processed through the internal 10-bit Analog-to-Digital Converter (ADC) of the ATmega328P.

The system operates using Timer1 configured in Clear Timer on Compare (CTC) mode. Through the Compare Match A interrupt vector (`__vector_11`), the microcontroller automatically performs periodic water quality evaluations without blocking the main execution flow. Every interrupt cycle triggers a complete sensor acquisition and processing routine.

During execution, the firmware initiates an ADC conversion by enabling the ADSC bit and reads the resulting 10-bit value from the ADCL and ADCH registers. The raw sensor value is then processed using inline arithmetic operations and fixed-point mathematical scaling to estimate Nephelometric Turbidity Unit (NTU) values. The implementation includes iterative 16-bit subtraction and overflow handling to support measurements greater than 999 NTU.

The calculated turbidity level is classified into three distinct water quality states:

- CLEAN → Low turbidity and safe water conditions
- NORMAL → Moderate turbidity levels
- DIRTY → High turbidity and contaminated water conditions

For visual feedback, the system uses a three-LED status matrix:

- the Clean LED indicates safe conditions,
- the Normal LED indicates moderate conditions,
- and the Dirty LED warns users about critical contamination levels.

The system also integrates a 16x2 LCD operating in 4-bit parallel mode. Instead of using I2C communication, the firmware directly controls the LCD through split nibble transmission (`send_nibble`) using the RS, EN, and D4–D7 data lines. This approach demonstrates efficient low-level bus management and direct hardware manipulation.

Additionally, the firmware streams real-time telemetry data through the UART interface at 9600 baud. The NTU values are dynamically converted into ASCII characters and transmitted in formats such as:

- NTU: 245
- Status: CLEAN
- Status: DIRTY
- > 999 (overflow condition)

The entire implementation emphasizes optimized bare-metal programming techniques, precise peripheral control, interrupt-driven scheduling, and efficient resource management within the ATmega328P architecture.

CHAPTER 3

TESTING AND ANALYSIS

3.1 TESTING

The testing of The DuckSaver system was carried out using the Proteus VSM simulation software to validate the implementation of the updated firmware and the modified hardware layout. The virtual environment integrated an ATmega328P microcontroller connected to an analog turbidity sensor at pin A0 (PC0), a 16x2 character LCD operating via a direct 4-bit parallel interface, a multi-tier LED status indicator array, and a UART Virtual Terminal.

The parallel LCD configuration was verified by mapping the Register Select (RS) pin to PB0, the Enable (EN) pin to PB1, and the data lines D4–D7 to pins PD4–PD7. The status indicators were monitored across three dedicated outputs: PD4 for the Clean state, PD5 for the Normal state, and PB5 for the Dirty state. During testing, the simulated analog voltage output from the turbidity sensor probe was adjusted dynamically across its full range to verify the microcontroller's ability to sample data, compute Nephelometric Turbidity Units (NTU) using assembly math, and update the output peripherals under changing water clarity conditions.

3.2 RESULT

The simulation demonstrated complete functionality across all operational ranges, displaying accurate live calculations on the serial console and transitioning seamlessly between three distinct water quality tiers:

High Turbidity State (Contaminated/Dirty Water): When the input from the sensor reflected low voltage conditions (representing thick suspended particles), the system calculated high turbidity values, such as NTU: 893 and NTU: 817. In this state, the Dirty LED (PB5) turned on, the other indicator pins were driven low, and the 16x2 LCD displayed Water Quality: on the first line and Status: DIRTY on the second line.

Mid-Range Turbidity State (Normal Water): As the simulated water began to clear and the calculated values dropped into the intermediate threshold zone (moving through values like NTU: 545, NTU: 454, and NTU: 345), the firmware automatically updated the outputs. The Dirty LED turned off, the Normal LED (PD5) turned on, and the LCD adjusted its text matrix to read Status: NORMAL.

Low Turbidity State (Clean Water): When the sensor input reached prime clarity levels (e.g., NTU: 273 down to NTU: 091), the system entered its safe baseline mode. The Normal LED deactivated, the Clean LED (PD4) was driven high, and the LCD updated to show Status: CLEAN.

Sensor Overflow Handling: To test the limits of the system, the input was pushed past the maximum rated threshold of the conversion formula. The firmware successfully trapped the exception, preventing numerical rollover or register wrapping, and continuously outputted NTU: > 999 to the Virtual Terminal.

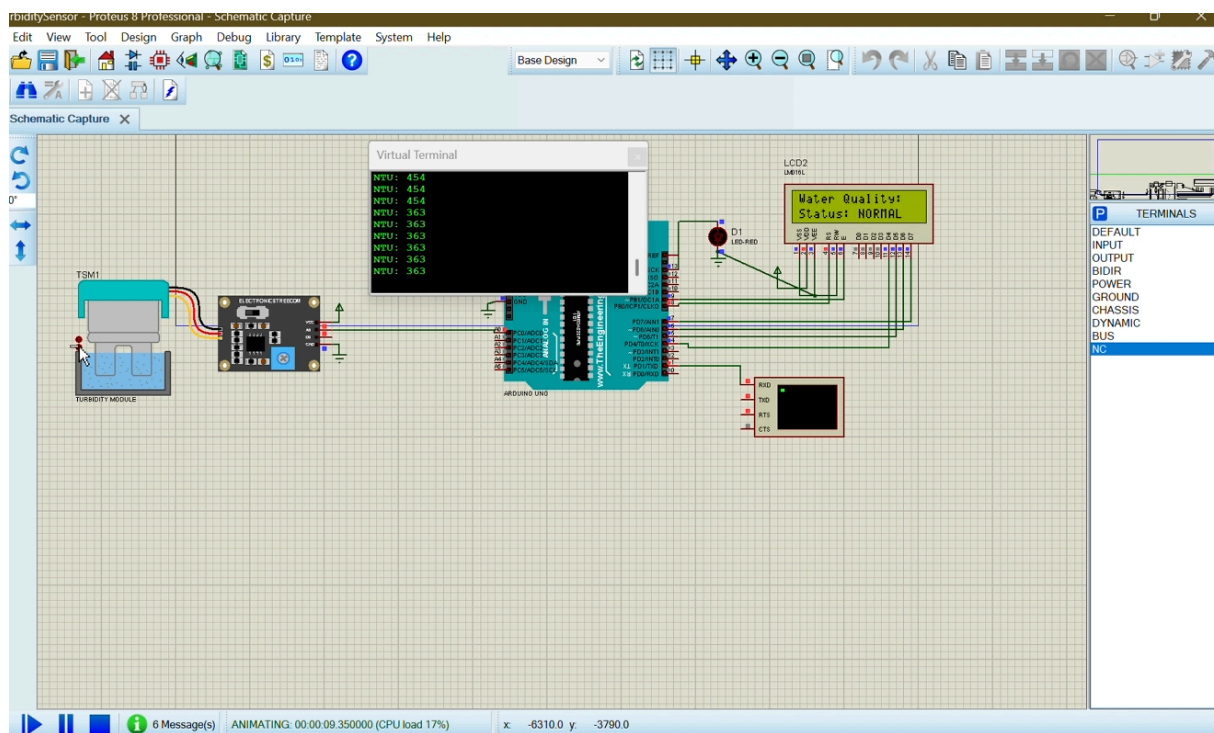
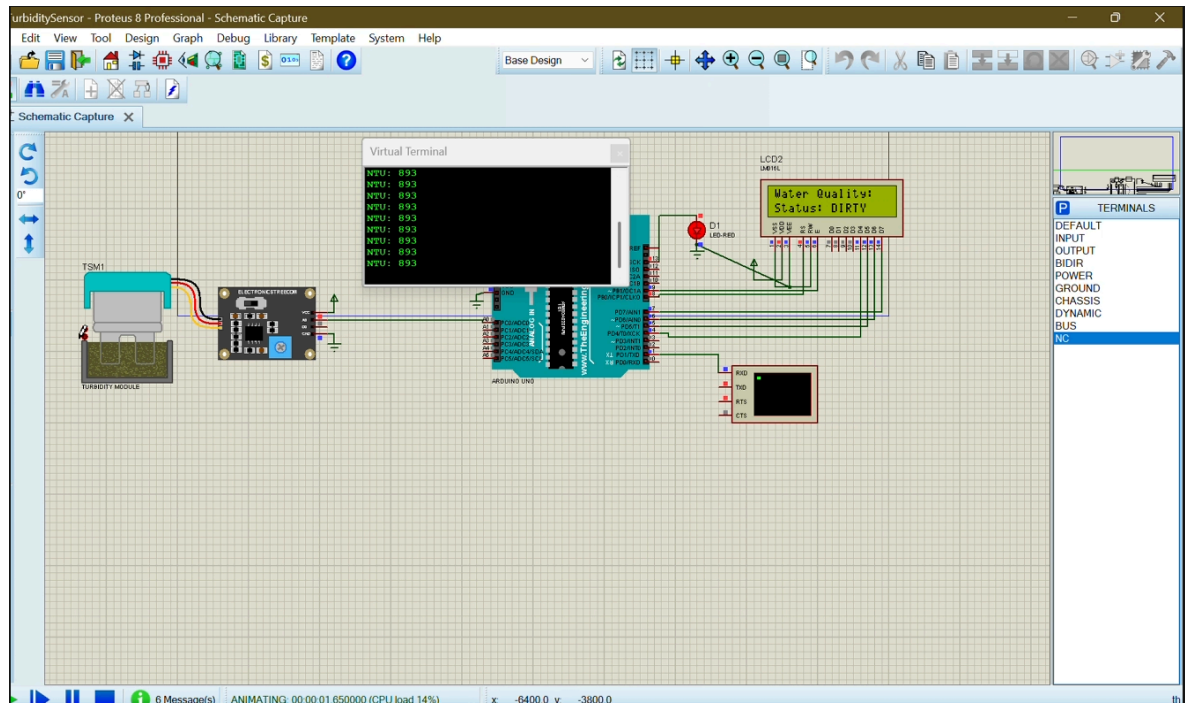


Fig 1. Testing Result

3.3 ANALYSIS

The updated implementation shifts away from external communication backpacks to a direct hardware control architecture, proving that multi-tier environmental monitoring can be performed efficiently using low-level programming.

The background operation is regulated by Timer1 configured in CTC mode. The initialization writes 0x7A to OCR1AH and 0x70 to OCR1AL alongside a 1024 clock prescaler (CS12 and CS10 set). This sets a precise, predictable timing cadence that forces execution into the Timer1 Compare Match A interrupt vector (`__vector_11`) where all sampling and logic occur, leaving the main loop unblocked.

Inside the interrupt service routine, the fixed-point arithmetic algorithm processes raw 10-bit data from ADCL and ADCH. The code subtracts the raw value from an empirical maximum sensor baseline constant of 819. If the result is positive, it uses coordinated logical left shifts (`lsl`) and rolling left shifts through carry (`rol`) to multiply the value across registers `r19:r20:r21`. The code then enters an iterative 16-bit division loop (`div_loop`), repeatedly subtracting a scaling factor of 614 while incrementing the counting registers `r16:r17` until the remaining dividend is smaller than the divisor. This approach successfully derives an integer NTU value entirely within assembly registers.

The logic tree uses multi-stage conditional comparisons (`cpi`, `tst`, `brsh`, `brlo`) to branch the execution path based on the final computed NTU values:

- Values above the upper limit jump to `status_dirty`.
- Values falling between the intermediate boundaries are routed to `status_normal`.
- Low values branch to `status_clean`.

The 4-bit parallel LCD subroutines show stable data bus performance. The `send_nibble` function reads the current state of `PORTD`, masks the lower control bits using an `andi` operation, merges the upper 4 bits of the target character byte using a logical `or`, and pushes the result back to `PORTD` before pulsing the Enable pin (`PB1`). The use of the `swap` instruction allows a single byte to be split into two sequential 4-bit nibbles, which updates the display rapidly and keeps character data clean during transitions.

CHAPTER 4

CONCLUSION

The DuckSaver project has successfully delivered a bare-metal, high-efficiency solution for real-time automated water quality tracking. By rewriting the system core in low-level AVR Assembly, the project demonstrates how direct micro-architectural management can replace high-level abstraction frameworks without sacrificing system capability.

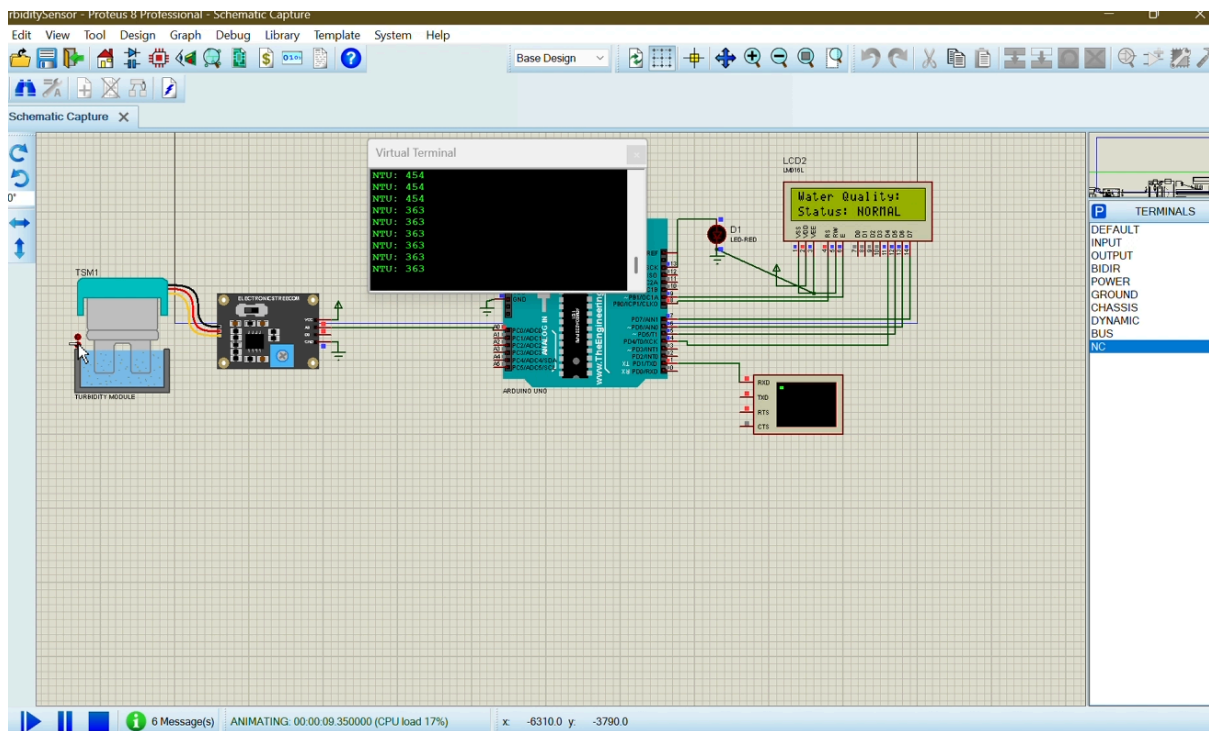
The updated firmware successfully integrates a multi-tier classification system (CLEAN, NORMAL, and DIRTY) by processing raw analog signals through an optimized 16-bit bit-shifting multiplication and division loop inside hardware timer interrupts. Furthermore, migrating the user interface to a direct 4-bit parallel display bus interface proves that an embedded system can achieve reliable, high-speed telemetry outputs and localized visual alerts using minimal hardware resources. Ultimately, *The DuckSaver* stands as an effective environmental monitoring device capable of tracking critical water turbidity parameters to help protect fragile aquatic habitats and support conservation efforts.

REFERENCES

- [1] D. Sahoo and A. Anandhi, “Conceptualizing turbidity for aquatic ecosystems in the context of sustainable development goals,” *Environmental Science: Advances*, vol. 2, pp. 1220–1234, 2023. [Online]. Available: <https://pubs.rsc.org/en/content/articlelanding/2023/va/d2va00327a>
- [2] R. Bilotta and R. Brazier, “Understanding the influence of suspended solids on water quality and aquatic biota,” *Water Research*, vol. 42, no. 12, pp. 2849–2861, 2008. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S0043135408001401>
- [3] W. F. Henley, M. A. Patterson, R. J. Neves, and A. D. Lemly, “Effects of sedimentation and turbidity on lotic food webs: A concise review for natural resource managers,” *Reviews in Fisheries Science*, vol. 8, no. 2, pp. 125–139, 2000. [Online]. Available: <https://research.fs.usda.gov/treesearch/1617>

APPENDICES

Appendix A: Project Schematic



Appendix B: Documentation

documentation.md

